Ali Nawaf

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Education

Case Western Reserve University | Cleveland, OH

MS in Machine Learning and Artificial Intelligence. May 2027

BS in Computer Science, Secondary Major: Mathematics. May 2027

Coursework: Intro to AI, Intro to ML, Networks, Security, HPC, Stochastic Calculus, Probability, Algorithms, Linear Algebra

Experience

AI Engineering Intern | Trek Health | San Ramon, CA Dec 2025 – Aug 2026

- Engineered a Python Lang Graph orchestrator in Snowflake to replace legacy stored procedures, enabling parallel sub-agent execution that reduced workflow latency by 37.5% and slashed LLM token consumption by 74.4%.
- Automated document classification, metadata extraction, stream ingestion and document caching into snowflake tables enriching the knowledge base, with a production scale of 8 million chunks for thousands of active users.
- Boosted RAG recall from 50% to 94% on a chosen test dataset, by building an agentic framework featuring multi-hop queries, query transformation, corrective rag and metadata enrichment to snowflake's cortex search.

ML Research Assistant | Houston Methodist Hospital | Houston, TX Aug 2025 – May 2026

- Developed ResNet50 deep learning models using MIMIC-IV and PTB-XL datasets to estimate cardiovascular age and assess structural cardiac health via standard 12-lead ECG and 3D Vectorcardiogram (VCG) modalities.
- Discovered VCG-derived age gaps as a superior ICU mortality biomarker (HR=1.22) compared to standard ECGs, utilizing Cox Proportional Hazards models, and Captum for feature attribution.

AI & SWE Intern | Heads-up Hockey | Waterford, VA Dec 2024 – Aug 2025

- Reduced manual annotation time by 70% by automating an annotation pipeline utilizing the SAM2 model.
- Doubled the active user base upon release by engineering Heads up Crush, an interactive (Duolingo like) Swift UI educational game featuring optimized on-device AI inference.
- Worked in a three engineers' team to improve the company's Hockey stick handling ML model's performance in production to more than 90% with fault detection and reporting.
- Engineered an end-to-end AI system by implementing data pre- and post-processing to extract goal counts, speed, and trajectory in Hockey shot practice, achieving 85% accuracy using the iphone/ipad camera.

AI Engineering Intern | Eaton | Cleveland, OH Jan 2025 – May 2025

- Engineered an automated Azure pipeline processing 10,000 monthly Salesforce service records.
- Developed Power BI call analytics and built an Agentic RAG chatbot for product documentation.

Projects

Ali Jr | Next.js, React, LiveKit, Google Cloud (Vertex AI / Gemini STT-TTS), LlamaIndex, ChromaDB, Docker, AWS
Hosted on AWS a voice agent digital avatar with a knowledge base that has access to all my classes, submissions, research and projects that I use on a daily basis to help me brainstorm and recall specific details related to old experiences.

Dispatch: Unified AI Agent Orchestrator Platform | Next.js, FastAPI, Supabase, Tauri, Python, TypeScript, Twilio
Built a voice-to-terminal orchestrator that executes local code via voice intent, integrating FastAPI and Twilio to convert voice commands into local terminal actions.

AI Medical Labeler | PyTorch, Python, API, git, GitHub, Docker, Networking
Reduced manual image annotation time by 60% by developing a Python-based tool that integrated with the Label-Studio API to pre-label images using SAM2 model.

Skills

Languages: Python, Java, SQL, Swift, JavaScript (Node.js, Next.js), TypeScript , C.

ML & AI: PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, Keras, Retrieval Augmented Generation (RAG), AI agents.

Cloud & DevOps: Azure, AWS, Power BI, SLURM, HPC, Kubernetes, AWS, Lamda functions, Linux, REST APIs